# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

# BIOLOGY

Past Paper 2024 • Paper Code: 2HA-I 24006 — 40061 (Version B) / 42061 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

## VERSION B

## BIOLOGY HSSC-II — SECTION – A (Objective Type)

**Time Allowed:** 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 40061 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Rapid melting of ice caps and glacier is due to:

(A) Greenhouse effect
(B) Ozone layer depletion
(C) Acid rain
(D) Algal bloom

Q2. Pressure filtration in kidney specifically occurs in:

(A) Glomerulus
(B) Bowman's Capsule
(C) Loop of Henle
(D) Vasa Recta

Q3. The lungs are covered by doubled layer thin membranous sacs called:

(A) Pleura
(B) Pericardium
(C) Peritonea
(D) Diaphragm

Q4. In the given diagram of skeletal muscle, bands i, ii and iii represent: (A labelled diagram of skeletal muscle is printed in the source scan.)

(A) i A band / ii I band / iii H-zone
(B) i I band / ii A band / iii Z-line
(C) i M-line / ii Z-line / iii H-line
(D) i H-line / ii I band / iii A band

Q5. Which of the following neurotransmitters function both as neurotransmitter and hormones? Describing perception of pain.

(A) Epinephrine
(B) Serotonin
(C) Dopamine
(D) Endorphins

Q6. After consuming banana, which hormone would be expected to increase in body?

(A) Prolactin
(B) Glucagon
(C) Insulin
(D) Parathyroid hormone

Q7. Which of the following hormones would bind to receptors located on the inside of a cell.

(A) Testosterone
(B) FSH
(C) Prolactin
(D) Somatotropin

Q8. The diagram shows female reproductive system in humans. In which labelled part is zygote formed? (A labelled diagram is printed in the source scan.)

(A) 1
(B) 2
(C) 3
(D) 2 and 3

Q9. Which of the following will be hemophilic?

(A) X^H X^H
(B) X^h Y
(C) X^H Y
(D) X^H X^h

Q10. In humans sex determination depends upon the nature of:

(A) Heterogametic female
(B) Homogametic female
(C) Heterogametic male
(D) Homogametic male

Q11. Which of the following shows evidences for evolution through molecular biology?

(A) Development of brain in vertebrates' embryo
(B) Distribution of species
(C) Comparison of gene/protein in different species
(D) Study of vestigial Organs

Q12. The cause of acid rain is:

(A) Oxides of carbon only
(B) Oxides of sulphur only
(C) Oxides of nitrogen and sulphur
(D) Oxides of nitrogen only

Q13. The enzyme used to isolate gene from DNA is:

(A) Helicase
(B) Reverse transcriptase
(C) Restriction endonuclease
(D) DNA polymerase

Q14. The agent which seperates the two strands of DNA in PCR (polymerase chain reaction) is:

(A) DNA ligase
(B) Primer
(C) Helicase
(D) Heat

Q15. Which of the following hormones is secreted if blastocyst is successfully implanted in the wall of uterus?

(A) hCG
(B) Corticosteroid
(C) LH
(D) Progesterone

Q16. The construction of intricate web by spider is an example of:

(A) Insight leaning
(B) Instinct
(C) Latent learning
(D) Habituation

Q17. Phosphodiester bond is:

(A) P-O-C-P-O-C
(B) C-O-P-O-C
(C) C-O-P
(D) C-C-O-P

## BIOLOGY HSSC-II — SECTIONS B & C (Subjective Type)

**Time Allowed:** 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 24006- (B)

Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14x3=42)

(i) [2+1] Describe the structure of rib cage. Briefly explain the advantages of having some ribs that are not attached directly to sternum. — OR (2+1) — What is the function of nephron? How is the rate of blood filtration with in the Bowman's capsule affected if blood pressure in the afferent arteriole is decreased?

(ii) [03] People with type I diabetes do not produce any insulin. What effect would it have on cells and metabolisms if left untreated? — OR (03) — Identify and describe the three different processes involved in embryonic development.

(iii) [03] The given pedigree shows a particular trait. Analyse the pedigree and draw conclusion. (A pedigree chart is printed in the source scan.) — OR (03) — What are chromosomes and genes? How are they related?

(iv) [03] In Griffiths experiment, do you think the heat treatment that killed the bacteria also inactivated the bacterial DNA? Why or why not? — OR (03) — Differentiate between convergent and divergent evolution on the basis of inheritance of homologous and analogous structure.

(v) [1+2] Why is nasal breathing generally considered superior to mouth breathing? Also write down sequence of muscle contraction that takes place during inhalation and exhalation. — OR (03) — Compare Klinefelter syndrome with Turner's syndrome with reference to Karyotype, Cause and Symptoms.

(vi) [03] FSH and LH get their names from events of female reproductive cycle but they also function in males. How their functions are similar in female and male? — OR (03) — Briefly explain Darwin's theory of natural selection.

(vii) [03] Differentiate between A. Ammonification and denitrification B. Xerarch and hydrarch succession — OR (03) — Briefly explain Integrated disease management.

(viii) [03] Briefly describe the procedure for the construction of Genomic library. — OR (03) — Compare and contrast the sympathetic and para sympathetic nervous systems.

(ix) [03] Briefly describe the following terms: A. Genetic marker B. Genomics C. Genome maps — OR (03) — Local anaesthesia blocks the opening of sodium channels in the nerve cells. How this would affect the transmission of pain impulses? Explain briefly.

(x) [03] Why biomass present at one trophic level of an ecosystem decreases at higher trophic level? Explain briefly. — OR (03) — How is it possible for a child to have a blood group O if the parents have blood group A and B.

(xi) [03] Write down any three applications of DNA analysis. — OR (1+2) — What is meant by home gardening? List at least four benefits of home gardening?

(xii) [1+2] State Mendel's law of segregating. Make a cross between round seed and wrinkled seed pea plant. — OR (03) — What are the three major steps in sequencing of DNA?

(xiii) [1+2] What is epistasis? How is it different from dominance? — OR (03) — Briefly explain characterises of Growth, Distribution and Carrying capacity of a population.

(xiv) [03] Briefly explain the role of vaccination as an effective method of preventing infectious diseases. — OR (03) — A plant with yellow flower was crossed with a plant with red flowers. The F1 progeny obtained had orange flowers. What is the inheritance pattern? Explain briefly.

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 2+4) Q.3 Identify the labelled parts and correlate these with major events of foetus development in the first trimester. (A labelled foetus diagram, parts A, B, C, D, is printed in the source scan.) — OR (2+4) — Identify the labelled parts E, F, G and H. Write down the major events of menstrual cycle in human females. (A menstrual-cycle graph, parts E, F, G, H, is printed in the source scan.)

Q4. (Marks 07) Q.4 Explain Habituation, Conditioning, Latent learning and insight learning by giving examples from human behaviours. — OR (5+2) — How is the information present in DNA used for the synthesis of RNA? Explain in detail. Also list down the post transcriptional modifications of mRNA.

Q5. (Marks 2+4) Q.5 What are the two methods used for gene therapy? Explain the role of successful gene therapy for cystic fibrosis. — OR (1+2+3) — Why anterior lobe of pituitary gland is called master gland? Enlist the hormones produced by anterior gland. Write down the functions of each hormones as well.

Q6. (Marks 07) Q.6 Describe the transport of oxygen and carbon dioxide through blood in humans. — OR (07) — Explain in detail the Sliding filament model of muscle contraction.

## VERSION D

### BIOLOGY HSSC-II — SECTION – A (Objective Type)

**Time Allowed:** 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 42061 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. The lungs are covered by doubled layer thin membranous sacs called:

(A) Pleura
(B) Pericardium
(C) Peritonea
(D) Diaphragm

Q2. The excretory product that requires maximum water for its removal is:

(A) Ammonia
(B) Creatinine
(C) Urea
(D) Uric acid

Q3. Which of the following is correctly matched? Sr./Feature/Bone/Cartilage: I Blood vessels (Bone ✓, Cartilage ×); II Living cells (Bone ✓, Cartilage ×); III Collagen (Bone ×, Cartilage ✓); IV Calcium phosphate (Bone ✓, Cartilage ×).

(A) I
(B) II
(C) III
(D) IV

Q4. During muscle contraction which of the regions shown in diagram decreases in length? (A sarcomere diagram with numbered regions is printed in the source scan.)

(A) 1 only
(B) 1 & 2 only
(C) 3 & 4 only
(D) 2,3 & 4 only

Q5. Which of the following condition best describes active membrane potential? (Charge-distribution diagrams are printed in the source scan.)

(A) +++ Outside / --- Inside
(B) +++ Outside / +++ Inside
(C) --+ Outside / --+- Inside
(D) --- Outside / +++ Inside

Q6. Which of the following hormones would be expected to increase, if a student reads all day and skips breakfast and lunch?

(A) Insulin
(B) Glucagon
(C) Secretin
(D) Calcitonin

Q7. Decrease of FSH and increase of estrogen, causes pituitary gland to secrete.

(A) Somatotrophic hormone
(B) Leutectrophic hormone
(C) Vassopressin
(D) Oxytocin

Q8. In given diagram of female reproductive system. The labelled parts are: (A labelled diagram is printed in the source scan.)

(A) I Ovary / II Fimbrae / III Oviduct / IV Vagina
(B) I Oviduct / II Ovary / III Uterus / IV Cervix
(C) I Uterine tube / II Ovary / III Cervix / IV Vagina
(D) I Uterine tube / II Fimbrae / III Cervix / IV Vagina

Q9. If a person has 44 autosomes +XXY, he will suffer form:

(A) Klinefelter's syndrome
(B) Turner's syndrome
(C) Down's syndrome
(D) Edward's syndrome

Q10. If CGATAG is sequence of nucleotide in DNA strand, the mRNA strand synthesized from it would be:

(A) GCUAUC
(B) GCTATG
(C) GCATAG
(D) CGUTUC

Q11. What limits the number of trophic level in a food chain?

(A) Primary productivity
(B) Efficient energy conservation
(C) Net productivity
(D) Species diversity

Q12. Which enzyme is used to join the desired gene into plasmid DNA during genetic engineering?

(A) DNA helicase
(B) DNA polymerase
(C) DNA ligase
(D) Taq polymerase

Q13. Which of the following enzymes is temperature insensitive?

(A) DNA Pol I
(B) DNA Pol III
(C) Taq Polymer
(D) RNA Polymer

Q14. The fluid which surrounds embryo is called:

(A) Chorionic fluid
(B) Amniotic fluid
(C) Uterine fluid
(D) Yolk

Q15. Working out a mathematical problem is an example of:

(A) Reflex
(B) Innate behavior
(C) Classical conditioning
(D) Insight learning

Q16. Large population size, random mating, no mutation and no migration are the postulates of:

(A) Theory of special creation
(B) Mendel's law of Segregation
(C) Law of independent assortment
(D) Hardy Weinberg's theorem

Q17. The particular array of chromosomes that an individual possesses is called:

(A) Genotype
(B) Phenotype
(C) Karyotype
(D) Wild type

## BIOLOGY HSSC-II — SECTIONS B & C (Subjective Type)

**Time Allowed:** 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 24006 (D)

Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14x3=42)

(i) [1+2] If people do not have ligaments, what would happen to them? Compare tendons with ligaments. — OR (03) — Explain the role of bacteria in nitrogen cycle.

(ii) [03] Complete the concept map by using the given terms: Neurons, Cell body, Neurotransmitter, Axon, Dendrites, Synapse. (A concept map diagram is printed in the source scan.) — OR (03) — Identify the organs A, B and C in the figure of the urinary system. Write function of each as well. (A urinary-system diagram is printed in the source scan.)

(iii) [03] Dead S-strain bacteria injected into mice did not harm the mice; but when mixed with living non-virulent R-strain bacteria, it killed them. Why? — OR (03) — A colour-blind father has a daughter with normal vision. The daughter marries a man with normal vision. What is the probability of her children to be colour-blind? Explain with the help of pedigree chart.

(iv) [03] Describe the path an oxygen molecule takes as it travels from nose to body cell. (Enlist each structure of respiratory system through which it passes.) — OR (1+2) — What is gene mutation? Briefly explain the causes and symptoms of Sickle cell anaemia as an example.

(v) [2+1] What are the functions of male reproductive system? What might happen to sperm production if a male has a high fever? — OR (03) — How is sewage treated? (Write down the three stages of sewage treatment process briefly.)

(vi) [03] Why is an embryo most vulnerable to drugs and other harmful substances taken by mother if it is between 2 to 7 weeks old? Also state how a foetus gets nourishment from mother. — OR (03) — Differentiate between: I Nucleosome & Primosome II Heterochromatin & Euchromatin III Sense codon & Non Sense codon.

(vii) [03] How would the inheritance pattern of an X linked dominant trait be different from that of X linked recessive trait? — OR (03) — "Migration may increase or decrease the effect of selection." Comment on the statement.

(viii) [03] How does inheritance of homologous and analogous structures result in convergent and divergent evolution? — OR (1+2) — What is animal husbandry? Briefly state the role of livestock in national economy.

(ix) [03] Discuss the role of microbes in industrial production. — OR (03) — Draw a flow chart to show how hormones control the function of male reproductive cycle.

(x) [03] Briefly explain the process of gene amplification through PCR (Polymerase Chain Reaction). — OR (03) — Briefly explain polygenic inheritance.

(xi) [03] Compare the anchorage dependent and anchorage independent techniques of animal cell culture. — OR (03) — How inhaled air in the respiratory tract is cleaned before it reaches the lungs? Describe briefly.

(xii) [1+2] Briefly explain haematoma formation. How is callus made during repair of bone structure? — OR (1+2) — What is drug addiction? Write down any two effects of drug addiction on central nervous system (CNS).

(xiii) [03] How kidneys help in maintaining homeostasis in the body? (Write role of kidney briefly) — OR (03) — Briefly explain the principle and process of Gel electrophoresis.

(xiv) [03] Describe the antagonistic effect of insulin and glucagon. — OR (03) — Briefly explain any three methods of plant breeding for crop improvement.

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 2+4) Q.3 The diagram shows how neurons communicate. Name the process and identify the parts 1,2 & 3. Explain the whole process in detail. (A neuron diagram is printed in the source scan.) — OR (1+5) — What is meant by innate behavior? Describe different types of orientation and non-orientation behavior with examples.

Q4. (Marks 2+4+1) Q.4 The diagram shows the neurulation in human embryo. Identify the labelled parts and state events of neurulation. Enlist the structures derived from neural crest cells. (A labelled neurulation diagram is printed in the source scan.) — OR (07) — Describe the components of recombinant DNA technology under the

following headings: (i) Gene of interest (ii) Molecular scissor (iii) Molecular glue (iv) Molecular carrier (v) Expression system.

Q5. (Marks 06) Q.5 Why multiple alleles provide many different phenotypes for a trait? Explain with the help of ABO blood group system. — OR (2+4) — Explain ecological succession. Write in detail about kinds of succession.

Q6. (Marks 5+2) Q.6 Describe the mechanism of breathing in humans. How is breathing controlled? Explain in detail. — OR (3+4) — Discuss the concept of genetic drift. Also describe speciation and different modes of speciation.